

System Analysis Document

Smart Study Planner Web Application

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Date: April 2026

Introduction

In light of the continuously increasing academic demands and the growing challenges students face in balancing coursework, deadlines, and personal well-being, effective time management has become a critical factor in academic success. Traditional study planning methods often fail to adapt to students' changing schedules, productivity patterns, and stress levels, leading to procrastination, burnout, and decreased academic performance.

The **Smart Study Planner** is an AI-powered academic productivity platform designed to help students organize their academic responsibilities through intelligent scheduling, smart task management, productivity analysis, and adaptive study support. By integrating modern web technologies and AI-driven features, the system transforms complex academic workloads into structured, personalized, and manageable study plans.

This document presents a comprehensive analysis of the **Smart Study Planner** system, including functional and non-functional requirements, key features, user stories, system specifications, detailed scenarios, acceptance criteria, and future scalability considerations.

🌸 Target Users

The Smart Study Planner system is designed to serve multiple types of users, each with different roles and responsibilities within the platform.

User Type	Description	Main Permissions
Student	The primary user of the system who manages study schedules, academic tasks, productivity tools, and AI-powered study features.	Create study plans, manage tasks, join study rooms, track academic progress, receive AI recommendations, and use productivity tools.
Doctor / Instructor	Academic instructors who provide course-related guidance, important deadlines, and academic announcements for students.	Add course announcements, share important deadlines, provide study guidance, and monitor general student progress if authorized.
Administrator (Admin)	Responsible for managing the platform, users, permissions, and maintaining overall system performance and security.	Manage users, monitor system activity, control permissions, manage reported issues, and maintain the platform.
Guest Visitor	Non-registered users who explore the landing page and platform features before creating an account.	Browse public pages and view general platform information only.

🌸 First : Requirements

1.1 Functional Requirements (FR)

Functional requirements describe the specific behaviors, functions, and services the system must provide.

ID	Functional Requirement
FR-01	The system shall allow users to create a new account and log in using email and password.
FR-02	The system shall allow users to add courses, including course name, instructor name, and credit hours.
FR-03	The system shall allow users to create academic tasks (assignments, projects, exams) and specify their due dates and difficulty levels.
FR-04	The system shall automatically generate a customized daily and weekly study schedule based on available study hours, task priority, and difficulty level.
FR-05	The system shall allow users to break down large tasks (e.g., projects) into smaller, manageable subtasks.

FR-06	The system shall send automatic notifications via browser or email 24 hours, 6 hours, and 1 hour before task deadlines.
FR-07	The system shall display a central dashboard summarizing daily tasks and upcoming deadlines with countdown timers.
FR-08	The system shall allow users to track academic progress through charts and completion rate indicators for each course.
FR-09	The system shall support a Focus Mode (Pomodoro) including study session timers, break timers, and distraction-free notification blocking.
FR-10	The system shall allow users to update task status (In Progress / Completed / Overdue) and delete or modify tasks.
FR-11	The system shall automatically reschedule overdue or missed tasks to the next available time slots.
FR-12	The system shall allow filtering and viewing of tasks by course, priority, completion status, or due date.
FR-13	The system shall allow the user to input tasks and study plans using voice commands and automatically convert them into organizable text.
FR-14	The system shall analyze the user's completion rate and study hours to provide intelligent suggestions for improving productivity.
FR-15	The system shall detect signs of academic burnout or low activity and send alerts and suggestions to take breaks.
FR-16	The system shall allow the user to activate group study mode and create shared study rooms with other users.
FR-17	The system shall allow the user to set weekly and monthly goals and track their achievement percentage.
FR-18	The system shall support an "Emergency Plan" feature that automatically rearranges the schedule when exams are approaching or tasks accumulate.
FR-19	The system shall allow the user to track actual study hours per course and display periodic reports accordingly.
FR-20	The system shall allow the user to sync the academic calendar with external services such as Google Calendar.
FR-21	The system shall support Dark Mode and Light Mode with the ability to switch between them.
FR-22	The system shall allow the user to add notes and attached files to each task or course.

FR-23	The system shall support an intelligent AI Assistant to answer user queries related to the study plan and productivity.
FR-24	The system shall display weekly and monthly statistics on academic performance and adherence to the study plan.
FR-25	The system shall allow the user to restore deleted tasks within a specified time period.
FR-26	The system shall allow the user to rate their focus level or mood after each study session to improve future recommendations.
FR-27	The system shall support achievements and motivational rewards (Gamification) to increase engagement and commitment.

1.2 Non-Functional Requirements (NFR)

Non-functional requirements describe the quality of the system and its technical and operational constraints.

ID	Non-Functional Requirement	Category
NFR-01	The system must be responsive, with main page loading times not exceeding 3 seconds under normal network conditions.	Performance
	All animations and visual effects shall be lightweight and shall not impact application performance.	Performance
	The system shall consume minimal device resources and power during prolonged operation.	Performance
NFR-02	The application must be compatible with multiple operating systems (Windows, macOS, Linux) and support common browsers (Chrome, Firefox, Edge, Safari).	Compatibility
	The system interface shall be compatible with both RTL and LTR writing directions.	Compatibility
	The system shall support push notifications on browsers and mobile devices in the future.	Compatibility
NFR-03	The design must be responsive to work efficiently on desktops, laptops, and tablets.	Usability
	The system must be easy to use with a low learning curve so any student can start without prior training.	Usability

	The system shall support multiple languages, including Arabic and English.	Usability
NFR-04	User data and passwords must be encrypted using standard security protocols (e.g., HTTPS and bcrypt encryption).	Security
NFR-05	The system must be available at least 99% of the time during the semester.	Reliability
	The system shall provide periodic backups of user data to prevent data loss.	Reliability
NFR-06	The system must support a scalable database to accommodate increasing numbers of users in the future.	Scalability
	The system shall be capable of handling a large number of tasks and data without noticeable performance degradation.	Scalability
NFR-07	The user interface must be clear with a cohesive visual design that enhances the academic experience.	Aesthetics
NFR-08	User data must be stored securely in the database and not shared with third parties.	Privacy
NFR-09	The system must be maintainable and future-developable to support AI features and smart recommendations.	Maintainability
	The system shall support future integration with external AI services and APIs.	Maintainability
	Errors and important system events shall be logged to facilitate maintenance and issue tracking.	Maintainability
NFR-10	The system interfaces shall comply with modern UI/UX standards.	User Interface Aesthetics

✿ Second: Key Features

Features describe what the system offers from the user's perspective, focusing on the practical benefits of each function.

Core Features

Feature 1: Customizable Academic Dashboard

Provides a central interface displaying a comprehensive summary of the daily schedule and upcoming tasks with countdown timers for deadlines, giving the student an immediate view of their daily priorities without needing to navigate between multiple pages.

Feature 2: Course & Subject Organizer

Enables students to add and manage their current courses, including lecturer name and credit hours, providing a centralized database for all registered subjects.

Feature 3: Smart Task Management

Allows users to create, categorize, prioritize, edit, and manage academic tasks (assignments, exams, projects), with support for breaking down large tasks into smaller achievable subtasks and automatically redistributing delayed tasks into future available study periods.

Feature 4: Automated Smart Scheduling

The system automatically generates customized daily and weekly study schedules based on available hours, task difficulty, deadlines, and priorities, helping students maintain balanced and efficient study routines.

Feature 5: Progress Tracking & Goal Visualization

Displays graphical indicators, progress charts, and achievement percentages for subjects, weekly goals, and monthly objectives, helping students monitor academic performance and maintain long-term motivation.

Feature 6: Automated Smart Alert System

The system sends automatic multi-stage reminders (24 hours, 6 hours, and 1 hour before deadlines) through browser notifications or email to ensure no assignment, project, or exam is missed.

Feature 7: Study Session Timer & Focus Mode

Provides an integrated Pomodoro-based focus system that manages study and break sessions, blocks distracting notifications, and offers a distraction-free study environment to improve concentration and study efficiency.

AI-Powered Features

Feature 8: Voice Command Integration

Allows students to add tasks and study plans using natural voice commands, which are automatically converted into organized and editable text, reducing manual input effort and improving accessibility.

Feature 9: Intelligent Productivity Advisor

Analyzes study hours, task completion rates, focus sessions, and productivity patterns to provide personalized recommendations that help students improve their study habits and time management.

Feature 10: Burnout Detection & Wellness System

Monitors prolonged study sessions, inactivity patterns, incomplete tasks, energy levels, and academic stress indicators to detect burnout risks. The system responds with supportive alerts, smart break recommendations, workload redistribution, and adaptive interface adjustments such as calming colors and simplified layouts during stressful periods.

Feature 11: Emergency Exam Planner

Provides a one-click emergency mode that instantly creates an intensive review schedule before exams by prioritizing critical topics and temporarily deprioritizing less urgent tasks.

Feature 12: Memory Weakness Analyzer

Identifies subjects or topics where the student frequently struggles or delays progress, then recommends targeted review sessions, alternative study techniques, and reinforcement plans to strengthen weak areas.

Feature 13: Deadline Risk Prediction

Analyzes task completion speed, remaining time, and current productivity levels to predict potential delays and warn users before deadlines are missed, while also suggesting intensive catch-up plans.

Feature 14: Energy-Based Dynamic Planning

Tracks virtual indicators such as focus level, energy level, and motivation level, then dynamically adjusts study schedules by reducing workload during low-energy periods and increasing recovery breaks when necessary.

Feature 15: AI Voice Motivation

Delivers short AI-generated motivational voice messages before or after study sessions, reading the daily plan aloud and encouraging students to stay committed and productive.

Feature 16: Anti-Procrastination Challenges

Detects procrastination patterns and low engagement levels, then introduces mini challenges, micro-goals, and motivational prompts that encourage students to return to productive study behavior.

Advanced & Collaborative Features

Feature 17: Group Study Rooms

Enables students to create or join shared virtual study rooms where they can collaborate with peers, exchange motivation, and maintain collective accountability while studying remotely together.

Feature 18: Productivity Heatmap

Displays an interactive heatmap visualizing the student's most productive days, peak focus hours, and academically weak periods, enabling data-driven scheduling decisions.

Feature 19: Immersive AI Study Room

Provides a fully immersive virtual study environment featuring ambient sounds (rain or café), lo-fi music, digital clocks, task lists, and futuristic neon visual effects to create a comfortable and distraction-free atmosphere for deep concentration.

❁ Third: User Stories

User stories are brief descriptions of user needs, written from their perspective, highlighting what they want and why.

US-01: Create Account and Login

As a new student,
I want to create an account in the system and log in,
So that I can access all application features and save my data securely.

US-02: Add and Manage Courses

As a university student,
I want to add my semester courses with their details,
So that I can organize my tasks by each subject separately and systematically.

US-03: Create Academic Tasks and Set Priorities

As a student,
I want to add my academic tasks (assignments, projects, exams) with their deadlines and difficulty,,
So that I can track my academic commitments and not miss any deadline.

US-04: Generate Automatic Customized Study Schedule

As a student who struggles with time management,
I want the system to automatically generate a daily study schedule that fits my available hours and task priorities,
So that I can follow a realistic study plan without manual planning.

US-05: Break Down Large Tasks into Subtasks

As a student with a large research project,
I want to break this project down into smaller subtasks distributed over multiple days,
So that I feel less pressure and complete the work gradually and organized manner.

US-06: Receive Smart Notifications Before Deadlines

As a busy student,
I want the system to send me automatic notifications well before deadlines,
So that I don't forget any important date and can prepare in time.

US-07: Track Academic Progress Graphically

As a student keen to improve my academic level,
I want to see clear charts showing my completion rates for each subject,
So that I can evaluate my performance and identify areas needing more effort.

US-08: Use the Study Timer (Pomodoro)

As a student who suffers from distractions while studying,
I want to use a Pomodoro timer to organize focus sessions and break periods,
So that I can study more efficiently and avoid mental exhaustion.

US-09: Add Tasks Using Voice Commands

As a student who is often busy while studying,
I want to add tasks and study plans using my voice instead of typing manually,
So that I can save time and interact with the system more easily.

US-10: Analyze Productivity and Receive Smart Suggestions

As a student who wants to improve study performance,
I want the system to analyze my study hours, productivity patterns, task completion rates, and peak focus times, then provide intelligent recommendations,
So that I can improve my study habits and manage my time more effectively.

US-11: Detect Academic Burnout and Receive Wellness Support

As a student who studies for long hours,
I want the system to detect signs of academic burnout or stress and provide supportive alerts and break recommendations,
So that I can maintain my mental well-being and avoid exhaustion.

US-12: Automatically Reorganize Schedule After Delays

As a student who may occasionally fall behind on tasks,
I want the system to automatically reorganize my study schedule when I miss deadlines or study sessions,
So that I can continue following my plan without losing organization.

US-13: Generate Emergency Exam Preparation Plans

As a student who with an upcoming exam and limited preparation time,
I want the system to instantly generate an intensive revision plan,
So that I can focus on the most important topics before the exam.

US-14: Join Collaborative Study Rooms

As a student who enjoys studying with others,
I want to join collaborative virtual study rooms,
So that I can stay motivated and study together with peers remotely.

US-15: Receive AI Motivational Messages

As a student who sometimes loses motivation while studying,
I want the system to send me motivational voice or text messages during study sessions,
So that I feel encouraged and remain committed to my academic goals.

US-16: Overcome Procrastination During Study

As a student who occasionally procrastinates,
I want the system to provide mini challenges and small achievable goals when my productivity decreases,
So that I can gradually return to focused and productive study behavior.

US-17: View Productivity Heatmap and Focus Insights

As a student who wants to better understand my study habits,
I want to view a productivity heatmap showing my most productive study hours and weak periods,
So that I can organize my schedule based on my peak focus times.

US-18: Experience an Immersive AI Study Environment

As a student who needs a comfortable and distraction-free study atmosphere,
I want the system to provide immersive study environments with ambient sounds, lo-fi music, and focus-friendly visual effects,
So that I can concentrate deeply and enjoy a more productive study session..

❁Fourth: Scenarios

Scenarios provide a detailed, step-by-step illustration of how the user interacts with the system in specific real situations.

Scenario 1: Register New Account and Login

User: Sarah, a new university student using the platform for the first time.

Goal: Create a new account and securely access the Smart Study Planner platform.

Steps:

- 1) Sarah opens the Smart Study Planner website.
- 2) The system displays multiple authentication options:
 - i. Create a new account
 - ii. Login with email and password
 - iii. Continue with Google
 - iv. Continue with GitHub
- 3) Sarah chooses to create a new account using her email.
- 4) The system displays the registration form.
- 5) Sarah enters her full name, email address, and password.
- 6) The system validates the entered information.
- 7) The account is successfully created.
- 8) Sarah logs into the platform using her credentials.

- 9) The system verifies the login information.
- 10) Sarah successfully accesses her personalized dashboard.

Alternative Flow:

Sarah may directly log in using her Google or GitHub account without manual registration.

Scenario 2: Create Courses and Organize Academic Subjects

User: Ahmad, a second-year computer science student

Goal: Add his university courses to organize his study plan efficiently

Steps:

- 1) Ahmad logs into the Smart Study Planner dashboard.
- 2) The system displays the main dashboard with an option “Add New Course”.
- 3) Ahmad clicks on “Add Course”.
- 4) The system shows a form requesting course details.
- 5) Ahmad enters:
 - i. Course name: Data Structures
 - ii. Instructor name: Dr. Khaled
 - iii. Credit hours: 3
- 6) The system validates the input data.
- 7) The course is successfully saved and added to his course list.
- 8) The system displays the updated list of courses on the dashboard.
- 9) Ahmad can now assign tasks under this course.

Alternative Flow:

Ahmad may edit or delete a course later if the semester schedule changes

Scenario 3: Create Academic Task and Set Deadline

User: Lina, a university student

Goal: Add an assignment and track its deadline

Steps:

- 1) Lina opens the Smart Study Planner dashboard.
- 2) She selects the course “Web Development”.
- 3) She clicks on “Add New Task”.
- 4) The system opens a task creation form.
- 5) Lina enters:
 - i. Task name: Project 1 - Website Design
 - ii. Type: Assignment
 - iii. Due date: 25/05/2026
 - iv. Difficulty level: High
- 6) The system saves the task successfully.
- 7) The task appears in the course task list and dashboard.
- 8) The system automatically includes it in the study schedule.

Alternative Flow:

Lina can later update the task or break it into subtasks.

Scenario 4: Automatic Study Schedule Generation

User: Omar, a student with multiple deadlines

Goal: Generate a smart study plan automatically

Steps:

- 1) Omar logs into the system.
- 2) The system analyzes his:
 - i. Courses
 - ii. Tasks
 - iii. Deadlines
 - iv. Available study hours
- 3) Omar clicks “Generate Study Plan”.
- 4) The AI scheduling engine analyzes task priorities, deadlines, study difficulty levels, and available study hours to generate an optimized personalized study schedule.

- 5) A personalized daily and weekly study schedule is generated.
- 6) The schedule is displayed on the dashboard in calendar view.
- 7) Omar reviews the plan and accepts it.
- 8) The system starts sending reminders based on the schedule.

Alternative Flow:

Omar can manually adjust time slots if needed.

Scenario 5: Focus Mode (Pomodoro Session)

User: Sara, a student preparing for exams

Goal: Study using focused sessions without distractions

Steps:

- 1) Sara selects a task from her dashboard.
- 2) She clicks “Start Focus Mode”.
- 3) The system activates Pomodoro timer (25-minute focus session).
- 4) Notifications and distractions are temporarily blocked.
- 5) A countdown timer appears on screen.
- 6) After 25 minutes, the system triggers a break notification.
- 7) Sara takes a 5-minute break.
- 8) The system repeats the cycle based on settings.

Alternative Flow:

Sara can pause or stop the session anytime.

Scenario 6: Smart Notification System

User: Yousef, a busy student

Goal: Receive reminders before deadlines

Steps:

- 1) Yousef has an upcoming assignment due tomorrow.
- 2) The system detects the deadline automatically.
- 3) The system sends notifications at:
 - i. 24 hours before
 - ii. 6 hours before

- iii. 1 hour before
- 4) Notifications appear on browser and email.
- 5) Yousef opens the system from the notification.
- 6) The task is highlighted in the dashboard.

Alternative Flow:

Yousef can snooze or disable notifications temporarily.

Scenario 7: Voice Command Task Entry

User: Reem, a student multitasking while studying

Goal: Add tasks using voice instead of typing

Steps:

- 1) Reem clicks the microphone icon in the system.
- 2) The system activates voice recognition mode.
- 3) Reem says: “Add assignment for AI course due next Monday.”
- 4) The AI voice recognition system converts the spoken command into structured and editable academic task data.
- 5) The task form is auto-filled.
- 6) The system asks for confirmation.
- 7) Reem confirms the task.
- 8) The task is saved successfully.

Alternative Flow:

If voice is unclear, the system asks for repetition or correction.

Scenario 8: Emergency Exam Planning

User: Ahmad, student with limited time before exams

Goal: Quickly generate revision plan

Steps:

- 1) Ahmad clicks “Emergency Plan” button.
- 2) The system analyzes all upcoming exams and tasks.
- 3) Less important tasks are deprioritized.
- 4) The system generates an intensive revision schedule.
- 5) High-priority topics are scheduled first.
- 6) Ahmad reviews the plan.
- 7) The plan is applied to his calendar automatically.

Alternative Flow:

Ahmad can manually adjust priorities.

Scenario 9: Burnout Detection and Alert System

User: Lina

Goal: Detect academic stress and prevent burnout

Steps:

- 1) The system monitors Lina’s study patterns.
- 2) The AI system analyzes continuous study duration, inactivity periods, missed breaks, task overload, and productivity decline.
- 3) It detects long study hours with low breaks.
- 4) Productivity level drops significantly.
- 5) The system identifies burnout risk.
- 6) An alert is displayed:
 - i. “You need a break”
- 7) The system suggests:
 - i. Rest time
 - ii. Light study plan
- 8) Lina accepts the recommendation.

Alternative Flow:

Lina can ignore alert and continue studying.

Scenario 10: Group Study Room

User: Omar

Goal: Study collaboratively with classmates

Steps:

- 1) Omar creates a study room.
- 2) The system generates a join link.
- 3) He invites classmates.
- 4) Students join the room.
- 5) Shared study session starts with chat and timers.
- 6) Members collaborate on tasks.
- 7) The system tracks group session time.

Alternative Flow:

Users can leave the room anytime.

Scenario 11: AI Study Friend Interaction

User: Salma, a student preparing for multiple exams

Goal: Receive emotional support, motivation, and intelligent academic guidance from the AI assistant

Steps:

- 1) Salma opens the Smart Study Planner dashboard.
- 2) The AI Study Friend greets her with a personalized message:
 - i. “Welcome back, Salma. Ready for today’s study goals?”
- 3) The AI assistant analyzes:
 - i. Upcoming deadlines
 - ii. Study progress
 - iii. Focus sessions
 - iv. Recent productivity
- 4) The assistant notices increased academic pressure.
- 5) The AI Study Friend suggests:
 - i. A lighter study session
 - ii. A short break
 - iii. Reorganized priorities

- 6) The assistant reads today's study goals aloud.
- 7) Salma starts a recommended focus session.
- 8) The AI companion remains available throughout the session for motivation and support.

Alternative Flow:

Salma may minimize or disable the AI assistant temporarily from the settings panel.

❁ Fifth: Specifications (Tasks)

Specifications (Tasks) describe the detailed implementation steps required to fulfill each User Story. Each user story is broken down into small, actionable and testable tasks that define how the feature will be built.

US-01: Create Account and Login

Tasks:

1. Design the registration and login UI forms.
2. Implement email and password input validation.
3. Integrate authentication system (email/password + Google/GitHub OAuth).
4. Encrypt user passwords using secure hashing (bcrypt).
5. Connect authentication system to the database.
6. Redirect user to dashboard after successful login.
7. Handle login errors (invalid credentials, missing fields).

US-02: Add and Manage Courses

Tasks:

1. Create database schema for courses.
 2. Develop "Add Course" form (course name, instructor, credit hours).
 3. Implement CRUD operations (Create, Read, Update, Delete) for courses.
 4. Display list of courses in dashboard.
 5. Allow editing and deleting courses.
 6. Link courses with related tasks.
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US-03: Create Academic Tasks

Tasks:

1. Design task creation form.
 2. Add fields (task name, type, deadline, difficulty, course).
 3. Store tasks in database.
 4. Categorize tasks by course and priority.
 5. Display tasks in dashboard and calendar view.
 6. Allow editing and deleting tasks.
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US-04: Generate Automatic Study Schedule

Tasks:

1. Collect user data (tasks, deadlines, available hours).
 2. Implement scheduling algorithm (priority-based distribution).
 3. Generate daily and weekly study plans.
 4. Display schedule in calendar UI.
 5. Allow manual adjustments by user.
 6. Save schedule updates in database.
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US-05: Break Down Large Tasks

Tasks:

1. Add “Subtasks” feature in task model.
 2. Create UI for adding subtasks under main tasks.
 3. Allow users to define subtasks with deadlines.
 4. Track progress of each subtask.
 5. Update main task progress automatically based on subtasks.
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US-06: Smart Notifications

Tasks:

1. Implement notification system (browser + email).
 2. Create scheduler for reminders (24h, 6h, 1h before deadline).
 3. Connect notifications with task deadlines.
 4. Send automated alerts to users.
 5. Allow notification settings customization.
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US-07: Track Academic Progress

Tasks:

1. Collect task completion data per course.
 2. Design progress dashboard UI.
 3. Implement charts (completion rate, performance graphs).
 4. Calculate progress percentage per course.
 5. Update charts dynamically.
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US-08: Focus Mode (Pomodoro Timer)**Tasks:**

1. Implement Pomodoro timer (25/5 cycle).
 2. Create focus session UI.
 3. Add break timer functionality.
 4. Block notifications during focus sessions.
 5. Track completed focus sessions.
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US-09: Voice Command Input**Tasks:**

1. Integrate speech-to-text API.
 2. Create voice input button.
 3. Convert speech into structured task data.
 4. Validate and confirm extracted task details.
 5. Save voice-generated tasks to database.
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US-10: Productivity Analysis System**Tasks:**

1. Collect user activity data (study hours, tasks, completion rate).
 2. Build analytics engine for performance evaluation.
 3. Generate productivity insights and reports.
 4. Display recommendations in dashboard.
 5. Update insights dynamically.
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US-11: Burnout Detection System

Tasks:

1. Monitor continuous study duration and inactivity patterns.
 2. Define burnout detection rules (low productivity, long sessions).
 3. Trigger alerts when burnout risk is detected.
 4. Provide break suggestions and workload reduction options.
 5. Log user response to alerts.
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US-12: Automatic Rescheduling

Tasks:

1. Detect overdue or missed tasks.
 2. Identify available time slots in schedule.
 3. Reassign tasks automatically.
 4. Update calendar dynamically.
 5. Notify user about schedule changes.
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US-13: Emergency Exam Plan

Tasks:

1. Detect upcoming exam deadlines.
 2. Prioritize critical tasks automatically.
 3. Generate intensive revision schedule.
 4. Adjust existing study plan.
 5. Display emergency plan in dashboard.
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US-14: Group Study Rooms

Tasks:

1. Create virtual study room system.
 2. Implement invitation and joining system.
 3. Add real-time chat or collaboration tools.
 4. Synchronize timers among participants.
 5. Track group session activity.
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US-15: AI Motivation System

Tasks:

1. Generate motivational messages using AI service.
 2. Trigger messages during study sessions.
 3. Support text and voice output.
 4. Personalize messages based on user behavior.
 5. Log user interaction with messages.
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US-16: Anti-Procrastination System

Tasks:

1. Detect inactivity or delayed tasks.
 2. Trigger mini-challenges or micro-goals.
 3. Display motivational prompts.
 4. Track response to interventions.
 5. Adjust difficulty of challenges dynamically.
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US-17: Productivity Heatmap

Tasks:

1. Collect study time data per hour/day.
 2. Generate heatmap visualization.
 3. Highlight peak productivity periods.
 4. Update heatmap in real time.
 5. Allow filtering by week/month.
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US-18: Immersive Study Environment

Tasks:

1. Design immersive UI (backgrounds, themes).
2. Integrate ambient sound system.
3. Add lo-fi music player.
4. Implement focus-friendly interface mode.
5. Allow user customization of environment.

❁ Sixth: Acceptance Criteria

Acceptance Criteria define the specific, measurable, and testable conditions that must be satisfied for each User Story to be considered complete and accepted by stakeholders. They act as a checklist to ensure that the implemented feature meets the required functionality and expected behavior.

US-01: Create Account and Login

ID	Acceptance Criteria
AC-01-1	The system must allow users to register using a valid email format.
AC-01-2	The system must reject duplicate email registrations with a clear error message.
AC-01-3	Passwords shorter than 8 characters must not be accepted.
AC-01-4	The system must support login using Google and GitHub accounts.
AC-01-5	User passwords must be securely encrypted using bcrypt or equivalent encryption methods.
AC-01-6	Users must be redirected to the dashboard after successful login.
AC-01-7	Invalid login attempts must display an appropriate error message.
AC-01-8	Login operations must be completed within 3 seconds under normal conditions.

US-02: Add and Manage Courses

ID	Acceptance Criteria
AC-02-1	Users must be able to add courses with course name, instructor name, and credit hours.
AC-02-2	Added courses must appear immediately in the dashboard.
AC-02-3	Users must be able to edit and delete existing courses.
AC-02-4	Duplicate courses with the same course code must not be allowed for the same user.
AC-02-5	Changes made to a course must automatically update related tasks.
AC-02-6	The system must display a confirmation message before deleting a course.

US-03: Create Academic Tasks

ID	Acceptance Criteria
AC-03-1	Users must be able to create tasks with title, deadline, difficulty level, and task type.
AC-03-2	Tasks must be linked to specific courses.
AC-03-3	Tasks without deadlines must not be accepted.
AC-03-4	Tasks must appear automatically in both dashboard and calendar view.
AC-03-5	Users must be able to edit, update, and delete tasks.
AC-03-6	Overdue tasks must be visually highlighted in red.
AC-03-7	Tasks must be sorted automatically based on priority and deadline.

US-04: Automatic Study Schedule

ID	Acceptance Criteria
AC-04-1	The system must generate a study schedule within 5 seconds after confirming available study hours.
AC-04-2	The generated schedule must not contain overlapping tasks or time conflicts.
AC-04-3	High-priority tasks must be scheduled before low-priority tasks.
AC-04-4	The system must consider task difficulty and deadlines while generating schedules.
AC-04-5	Users must be able to manually modify the generated schedule.
AC-04-6	Missed tasks must be automatically redistributed to future available slots.

US-05: Break Down Large Tasks

ID	Acceptance Criteria
AC-05-1	Users must be able to create subtasks under a main task.
AC-05-2	Each subtask must have its own deadline and completion status.
AC-05-3	Main task progress must update automatically according to completed subtasks.
AC-05-4	Subtasks must appear clearly in task details view.
AC-05-5	The system must prevent subtasks from having deadlines later than the main task deadline.

US-06: Smart Notifications

ID	Acceptance Criteria
AC-06-1	The system must send notifications 24 hours, 6 hours, and 1 hour before deadlines.
AC-06-2	Notifications must be delivered through browser and email.
AC-06-3	Notifications must be triggered automatically without manual user interaction.
AC-06-4	Users must be able to customize or disable notification settings.
AC-06-5	The system must maintain a notification history section.

US-07: Progress Tracking

ID	Acceptance Criteria
AC-07-1	The system must display graphical progress indicators for each course.
AC-07-2	Completion percentages must update automatically after task completion.
AC-07-3	Weekly and monthly productivity reports must be available.
AC-07-4	Courses with low completion rates must display visual warning indicators.
AC-07-5	Progress charts must support both percentage and numerical statistics.

US-08: Focus Mode (Pomodoro)

ID	Acceptance Criteria
AC-08-1	The system must provide a functional Pomodoro timer with focus and break cycles.
AC-08-2	Notifications must be blocked automatically during focus sessions.
AC-08-3	Users must be able to start, pause, and stop focus sessions.
AC-08-4	Focus session history must be stored and displayed in productivity reports.
AC-08-5	Users must be able to customize focus session duration between 15 and 60 minutes.
AC-08-6	The system must suggest a long break after four consecutive focus sessions.

US-09: Voice Command Input

ID	Acceptance Criteria
AC-09-1	The system must support voice-to-text task creation.
AC-09-2	Spoken commands must be converted into structured editable tasks.
AC-09-3	Users must be able to review and confirm detected voice input before saving.
AC-09-4	Voice processing must occur in real time without noticeable delay.

US-10: Productivity Analysis

ID	Acceptance Criteria
AC-10-1	The system must analyze study hours and task completion rates automatically.
AC-10-2	Personalized productivity recommendations must be generated dynamically.
AC-10-3	Recommendations must appear in the dashboard clearly.
AC-10-4	Productivity data must update continuously based on user activity.

US-11: Burnout Detection

ID	Acceptance Criteria
AC-11-1	The system must detect prolonged study sessions without breaks.
AC-11-2	Burnout alerts must be triggered when stress risk is detected.
AC-11-3	The system must recommend rest periods or lighter workloads.
AC-11-4	Burnout alerts must appear using clear and supportive language.

US-12: Automatic Rescheduling

ID	Acceptance Criteria
AC-12-1	The system must automatically detect missed or overdue tasks.
AC-12-2	Overdue tasks must be redistributed into future available time slots.
AC-12-3	The calendar must update immediately after rescheduling.
AC-12-4	Users must receive notifications about schedule updates.

US-13: Emergency Exam Plan

ID	Acceptance Criteria
AC-13-1	The system must generate intensive revision schedules for upcoming exams.
AC-13-2	High-priority topics must be scheduled before low-priority topics.
AC-13-3	Emergency plans must temporarily override regular schedules when necessary.
AC-13-4	Users must be able to review and accept generated emergency plans.

US-14: Group Study Rooms

ID	Acceptance Criteria
AC-14-1	Users must be able to create and join collaborative study rooms.
AC-14-2	Multiple users must be able to interact in real time داخل the same study room.
AC-14-3	Group study session duration must be tracked automatically.
AC-14-4	Users must be able to leave study rooms at any time.

US-15: AI Motivation System

ID	Acceptance Criteria
AC-15-1	The system must generate motivational text and voice messages.
AC-15-2	Messages must adapt based on user productivity and behavior.
AC-15-3	AI messages must not interrupt active focus sessions excessively.
AC-15-4	Users must be able to enable or disable AI motivation settings.

US-16: Anti-Procrastination System

ID	Acceptance Criteria
AC-16-1	The system must detect inactivity and low productivity patterns.
AC-16-2	Mini productivity challenges must be triggered automatically.
AC-16-3	Users must receive motivational prompts during inactivity periods.
AC-16-4	User responses to challenges must be tracked and stored.

US-17: Productivity Heatmap

ID	Acceptance Criteria
AC-17-1	The system must display a heatmap showing study activity levels.
AC-17-2	Peak productivity hours must be visually highlighted.
AC-17-3	Heatmap data must update automatically based on user activity.
AC-17-4	Users must be able to filter heatmap results by time range.

US-18: Immersive Study Environment

ID	Acceptance Criteria
AC-18-1	The system must provide ambient sounds and lo-fi music options.
AC-18-2	Users must be able to activate immersive focus mode manually.
AC-18-3	The interface must switch to a distraction-free layout during immersive mode.
AC-18-4	Users must be able to customize environment settings and sound preferences.

❁ Seventh: Software Architecture

The Software Architecture diagrams illustrate the overall structure of the Smart Study Planner system, including user interactions with the system functionalities and the physical deployment of software components across servers and external services.

Use Case Diagram

The Use Case Diagram represents the interactions between the different system users (actors) and the Smart Study Planner functionalities. It helps visualize the main services provided by the system from the user's perspective.

Main Actors:

- Student
- Instructor / Doctor
- Administrator (Admin)
- Guest Visitor

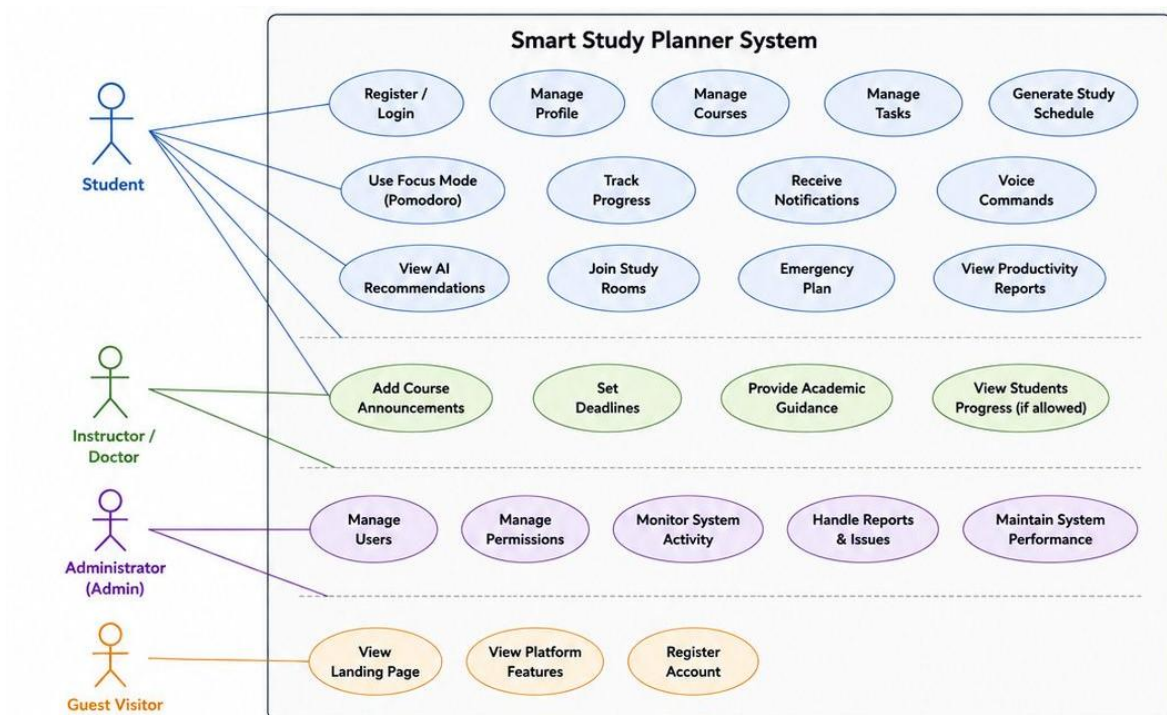
Main Use Cases:

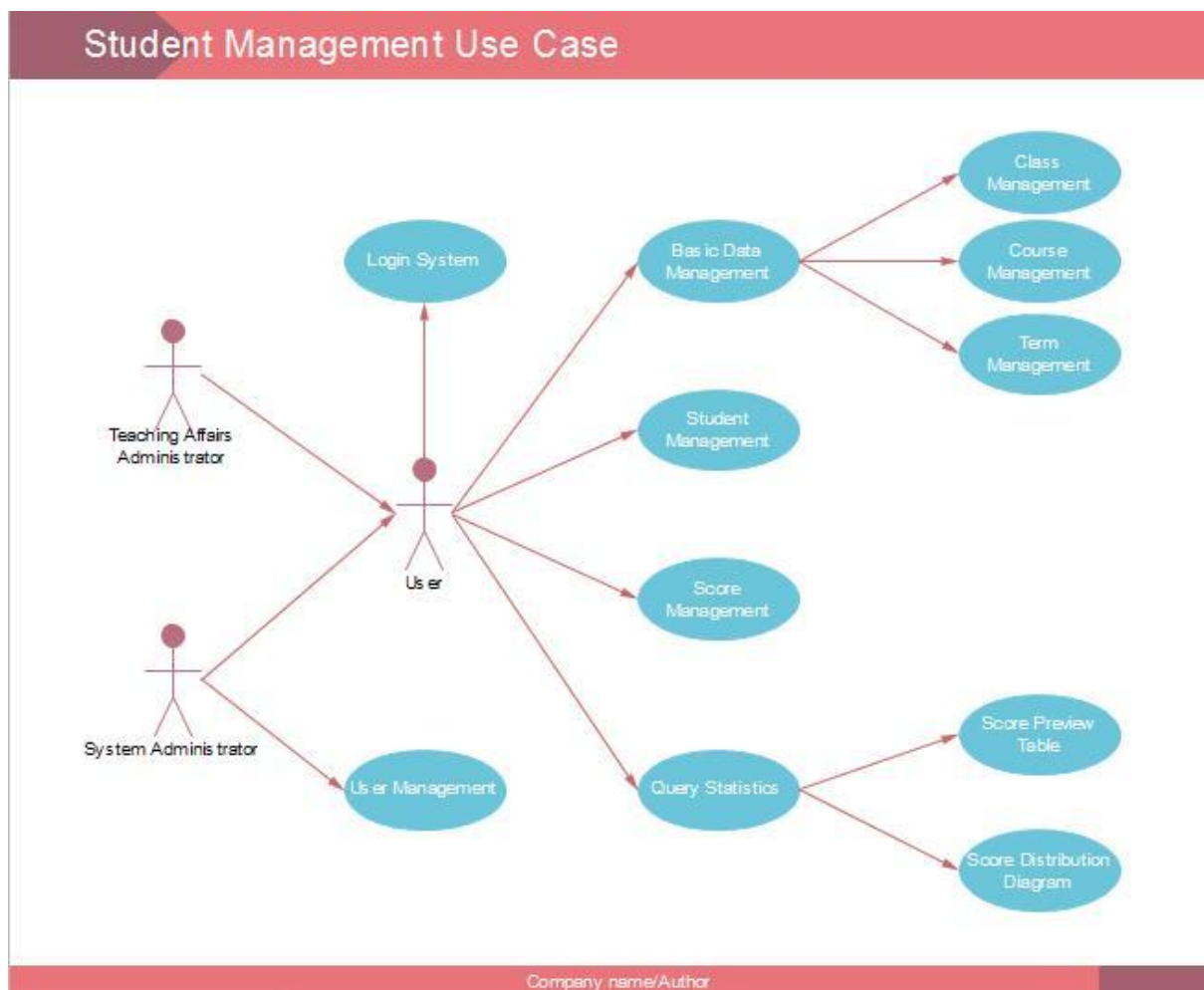
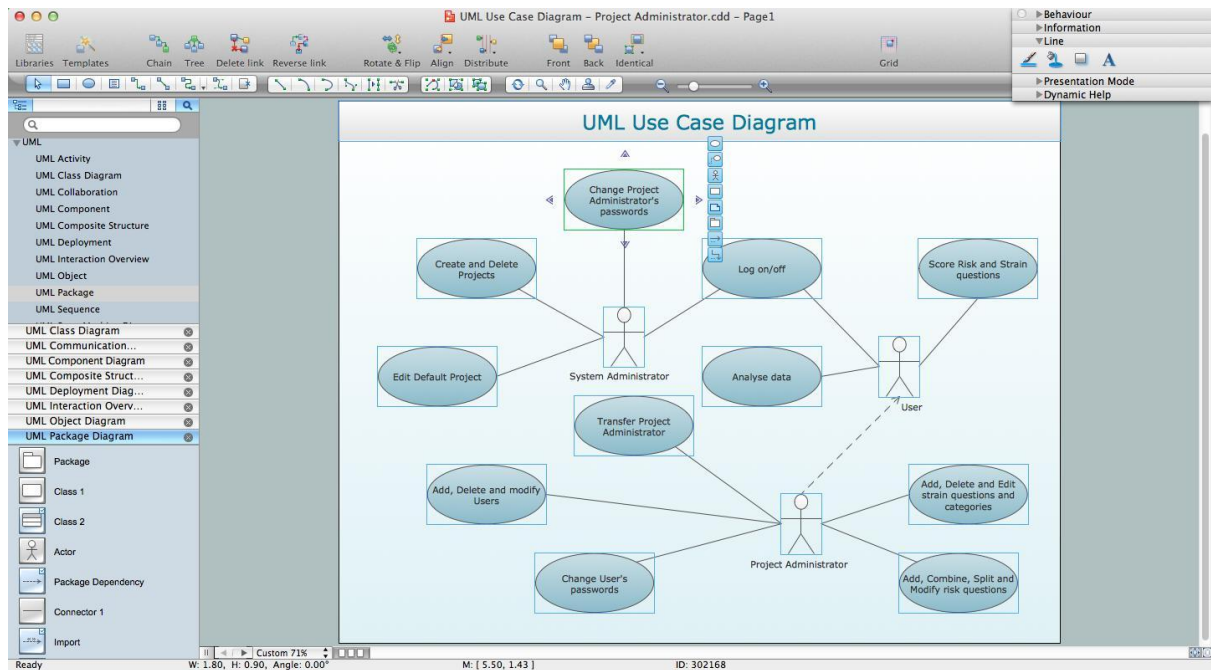
- Register and Login
- Manage Courses
- Manage Academic Tasks
- Generate Smart Study Schedule
- Use Focus Mode (Pomodoro)
- Receive Notifications

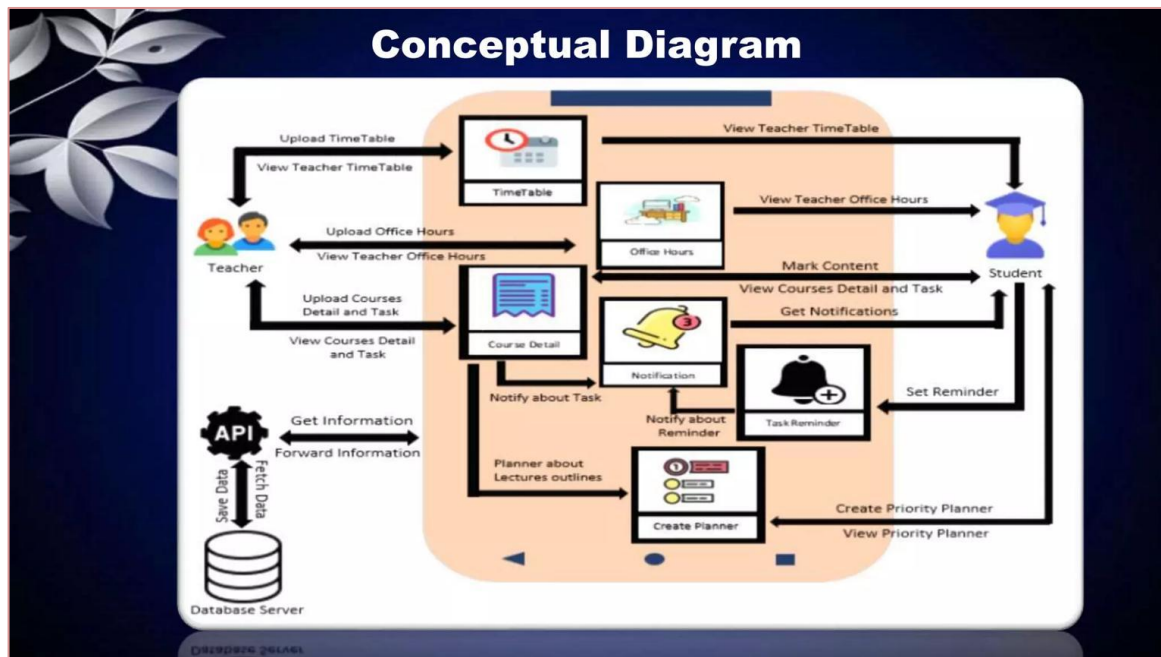
- View Productivity Reports
- Join Study Rooms
- Manage Users and Permissions
- View Platform Features

The Student is considered the primary actor because they interact with most system functionalities, while instructors and administrators are responsible for academic and administrative operations.

✿ Use Case Diagram Image







Deployment Diagram

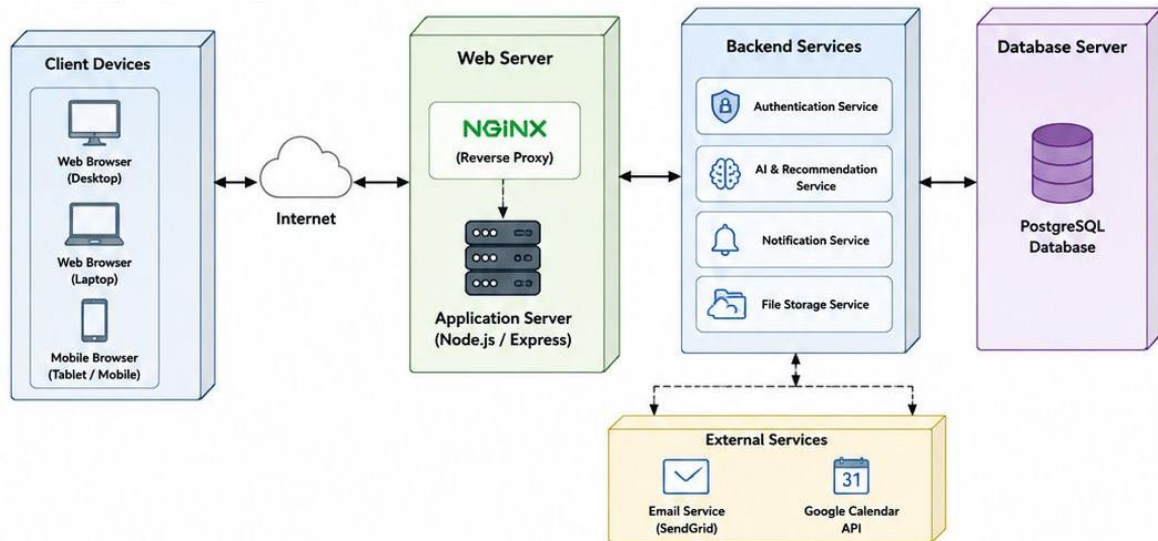
The Deployment Diagram illustrates how the Smart Study Planner system is physically deployed across different hardware and software components. It demonstrates the communication between client devices, web servers, backend services, databases, and external APIs.

Deployment Components:

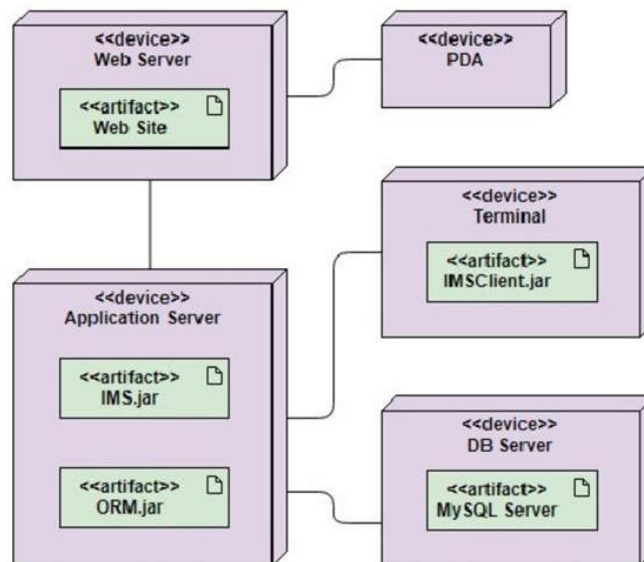
- **Client Devices:**
Desktop browsers, laptops, tablets, and mobile devices used by students and instructors.
- **Web Server:**
Handles incoming user requests and routes them to backend services.
- **Application Server / Backend Services:**
Responsible for authentication, AI recommendations, notifications, file storage, and business logic processing.
- **Database Server:**
Stores user accounts, tasks, schedules, reports, and academic data securely.
- **External Services:**
Includes email notification services and Google Calendar API integration.

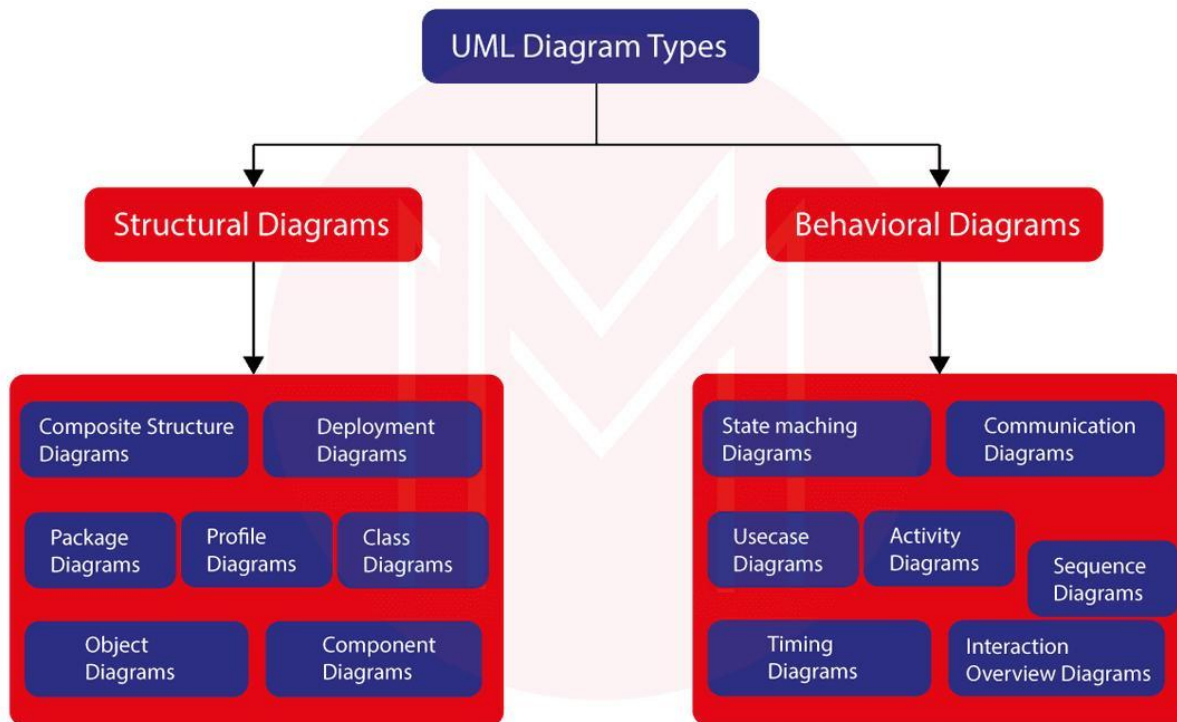
The architecture follows a scalable web-based client-server model that supports future expansion and AI integration.

Deployment Diagram Image



The system is a web application accessed through client devices via a web browser. User requests are sent over the internet to the Web Server, which forwards them to the Application Server. The backend services handle business logic, AI recommendations, notifications, and file storage. Data is stored securely in the PostgreSQL database. The system integrates with external services such as Email Service and Google Calendar API.





❁ Architecture Summary

The Smart Study Planner system is implemented as a modern web application using a multi-layer architecture. Users access the platform through web browsers, while backend services process business logic, AI Scheduling Engine, Voice Recognition Service, Productivity Analytics Engine, scheduling operations, and notifications. Data is securely stored in a centralized database, and the platform integrates with external services such as Google Calendar and email systems to improve productivity and usability.